

Improving UAV Aerial Imagery Detection Method via Superresolution Synergy

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Abstract—Unmanned aerial vehicles (UAVs) have emerged as versatile tools across various industries, providing valuable insights through aerial image analysis. However, the efficacy of UAV-deployed image detection systems is often limited by the resolution of captured images and the altitudinal constraints of UAV operations. This article introduces a novel integration of the detection system with superresolution networks and image reconstruction techniques, inspired by the exceptional visual capabilities of eagles, to enhance image detail and detection recall from aerial perspectives. The superresolution component utilizes advanced algorithms to upscale the resolution of images captured by UAVs, thereby improving the granularity and clarity of the visual data. Concurrently, image reconstruction techniques are applied to enhance the quality of original images further. In addition, we propose an innovative adaptive feature fusion technique, which not only surpasses traditional concatenation methods in integrating multiscale features effectively but also demonstrates remarkable improvement in feature utilization and further refinement of the fusion process. Extensive experiments conducted on VisDrone2019 and DOTA datasets demonstrate that our integrated system significantly outperforms existing methods in terms of detection precision and recall. Compared to YOLOv5s, recall and mAP50 have increased by 8.89% and 11.11%, respectively, with only a slight increase in the number of parameters and complexity.

Index Terms—Eagle-eye vision system, object detection, unmanned aerial vehicle (UAV) imagery, YOLOv5.

I. INTRODUCTION

THE rapid advancement of unmanned aerial vehicle (UAV) technologies has greatly expanded their range of applications, including agricultural monitoring, urban planning, and environmental conservation [1], [2]. At the core of these applications lies object detection, which allows UAVs to efficiently and accurately monitor ground activities. However, the effectiveness of image detection algorithms is often limited by the resolution of UAV sensors and the operational altitude of UAVs. As a

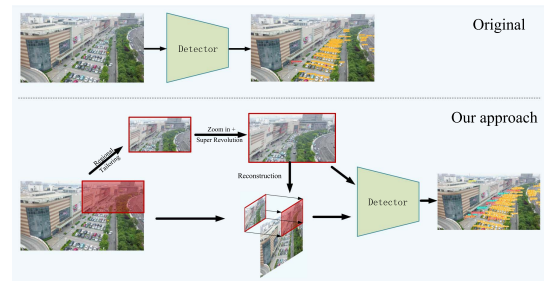


Fig. 1. Comparison of the methods in this article and traditional methods. The traditional approach involves feeding the image directly into the detector. Our proposed detector, which sends the localized map after SR to the detector along with the reconstructed map.

result, UAV-captured images frequently lack the detail needed for thorough analysis, particularly in identifying small or distant objects [3]. To address these challenges, the integration of superresolution (SR) techniques has proven effective in enhancing image details, enabling improved object detection, particularly for small or distant targets. Similar SR-based approaches have been explored in UAV imagery to mitigate limitations, such as low resolution caused by altitude or limited onboard camera quality. While SR techniques primarily focus on improving image quality, their deployment in UAV systems can benefit from efficient data handling strategies to ensure seamless integration and real-time processing [4], [5]. For instance, advancements in UAV communication frameworks and onboard processing capabilities could further support the adoption of SR-enhanced object detection in practical scenarios.

Drawing inspiration from the exceptional visual capabilities of eagles—combining high-resolution focus with a wide field of view [6]—this research introduces a novel approach to UAV image detection. Fig. 1 shows the difference between our method and conventional detectors. Traditional methods, which typically involve feeding entire images directly into detection systems, are effective for general use cases but often fall short in capturing the fine details necessary for high-precision tasks, especially when it comes to distant or small objects in aerial imagery. In contrast, our proposed method introduces a more sophisticated and targeted approach. Initially, specific regions of interest within an image are identified and cropped. These cropped sections are then subjected to an SR process, which enhances the image quality by increasing the resolution of the observed details while preserving the integrity of the original data. Following the SR enhancement, the high-resolution local

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patches are seamlessly reconstructed back into the original image framework, ensuring that the enhanced details are accurately integrated without disturbing the overall image composition. Finally, the reconstructed image, along with the high-resolution patches, is input into the detection system. We integrate the YOLOv5 detection system with SR network techniques to replicate an “eagle-eye” effect. This approach enhances both the precision and clarity of UAV-captured images. This synergy not only improves the granularity with which UAVs can detect features on the ground but also expands the operational range and accuracy of UAV surveillance systems. In addition, eagle eyes are equipped with unique anatomical features such as a double fovea and an exceptionally high number of photoreceptors, which allow them to discern fine details from extreme distances [7]. Drawing parallels to this natural phenomenon, the proposed system employs SR techniques to reconstruct high-resolution details from low-resolution images, thereby overcoming the traditional limitations posed by UAV imaging systems. Concurrently, the YOLOv5 object detection algorithm adapts these high-quality images to detect and classify objects with remarkable accuracy and speed.

The integration of SR and detection technologies is expected to set a new benchmark in UAV image detection. This article details the development and validation of this integrated approach, discusses its implications for UAV technology, and explores the future possibilities opened by this advancement. Through this innovation, UAVs may not only see as eagles do but also extend their vision beyond the natural capabilities of these avian predators. In summary, the main contributions of this article are as follows.

- 1) We introduce an innovative integration of the YOLOv5 object detection framework with SR networks to simulate the high-resolution, wide-field visual acuity of eagle eyes. This methodology significantly enhances the detail and accuracy of images captured by UAVs, facilitating more precise object detection even from high altitudes.
- 2) We propose a novel adaptive feature fusion (AFF) approach that dynamically integrates multiscale features from different network layers, tailored to the contextual demands of the aerial scene. This method provides a superior alternative to traditional concatenation techniques, optimizing feature utilization for improved detection performance.
- 3) Through extensive testing on standard aerial image datasets, our integrated system sets new benchmarks for detection precision and recall. The results indicate substantial improvements over existing detection methods, underscoring the effectiveness of combining detection with SR techniques.

II. RELATED WORK

A. Object Detection in UAV Imagery

In the realm of UAV aerial imagery, object detection faces unique challenges distinct from those encountered in traditional object detection. Specifically, the detection of small-sized targets

at long distances, coupled with significant variations in object scales, presents considerable difficulties.

In recent years, researchers have developed numerous methodologies to address these challenges. For example, Chen et al. [8] introduced a detector that combines an anchor-free approach with a regression module. This hybrid system utilizes the regression module to refine the initial rough bounding boxes generated by the anchor-free detector, resulting in more precise multiscale bounding box delineation. Li et al. [9] enhanced the detection accuracy of small targets by employing density maps to guide the cropping of dense regions in high-resolution aerial images. Each densely populated block of small targets was subsequently fed into the detector, optimizing the detection process for small objects. Yang et al. [10] developed a clustered detection network that integrates clustering methodologies within a single framework to enhance the detection of densely packed small objects adopting clustering techniques. Lu et al. [11] designed a hybrid method combining CNNs and transformers that utilizes multiscale information for more accurate UAV image detection. During the inference process, the method employs a fusion of original and sliced images to enhance the detection of small-scale objects. Guzman-Pando et al. [12] integrated the concept of the eagle’s deep fovea with detection technology, combining the magnifying ability of eagle eyes and the capacity to extract contextual information from peripheral vision. This approach excels in detecting small-sized targets. Zhang et al. [13] introduced the Drone-YOLO algorithm, incorporating the RepVGG block as the downsampling layer and integrating extra feature branches at the neck to enhance the detection accuracy of YOLOv8 on UAV imagery. Chen et al. [14] introduced a network that excels in UAV target detection by simultaneously enhancing network learning capabilities and feature strength through dynamic sample allocation and attention augmentation. Wu et al. [15] proposed the CDYL algorithm, which is based on fully connected deformable convolution and possesses the capability for adaptive feature aggregation. This algorithm efficiently captures both edge and contextual information, enhancing the detection of densely packed small targets. Yi et al. [16] developed the LAR-YOLOv8 algorithm, which employs transformer blocks to enhance feature extraction, and introduces a novel RIOUS loss function, markedly optimizing the detection of small-scale targets in remote sensing imagery (RSI). Jiang et al. [17] developed MFFSODNet, a network that enhances detection performance in UAV-captured images by fusing larger scale backbone features and deliberately sacrificing the detection of larger targets to improve precision for smaller objects. Fu et al. [18] proposed a network that fuses local and global features through hybrid branching, which adaptively exchanges complementary information to enhance the extraction of targets in RSIs.

B. SR-Assisted Object Detection

Whether it is through multiscale features [9], [19], hard negative mining [14], or context enhancement techniques [20], all have proven effective for improving the detection performance of small-sized targets. However, due to cost considerations, these

methods are predominantly applied to low-resolution images, often overlooking the rich details contained in high-resolution features. Nevertheless, the integration of SR with detection has been demonstrated to be effective in object detection tasks, effectively utilizing the detailed information provided by high-resolution features [21], [22].

Shermeyer and Van Etten [23] demonstrated the significant impact of SR on the detection capabilities of various resolutions and categories in satellite imagery by quantifying its effects, underscoring the importance of SR in satellite image detection. Ji et al. [24] employed generative adversarial networks to perform SR on remote sensing datasets, enabling an end-to-end network to accurately detect small-sized vehicles. Deng et al. [25] introduced a global-local adaptive network that utilizes global features and local information to precisely estimate bounding boxes for original downsampled and cropped maps. This network enhances the detection of small-scale targets through refined SR of locally cropped images. Courtrai et al. [26] input both the original and downsampled images into the detector, subsequently applying SR to the downsampled images to enhance the network's ability to detect small targets in remote sensing. Rabbi et al. [27] enhanced the quality of RSIs using a SR GAN network and improved the detection performance for small targets through the integration of residual dense blocks. Meethal et al. [28] enhanced the detection of small objects by developing an algorithm to identify and crop dense image regions. They improved the detector's performance by enlarging these cropped images to a higher resolution before feeding them into the detector. Zhang et al. [29] proposed a GAN-based SR method with a high-resolution feature extractor that enhances small-ship detection in optical RSIs by directly extracting high-resolution features from low-resolution images, improving performance while reducing computational costs. Zhang et al. [30] enhanced the detection accuracy of remote sensing targets by employing an SR network to augment and fuse multiscale features from the backbone. This approach not only learned features from low-resolution inputs but also preserved high-resolution information, thereby improving the network's detection capabilities. In addition, by discarding the auxiliary SR component during the inference phase, they achieved reduced complexity in the network.

III. METHODOLOGY

A. Baseline Architecture

In this article, we employ YOLOv5 as the baseline detector, whose main architecture is illustrated in Fig. 2 and comprises three primary components: the backbone, neck, and head. The backbone is responsible for progressively extracting increasingly complex features from the input images, which serve as the core information relied upon by both the neck and head. The neck utilizes a feature pyramid structure, which aids in the contextual fusion and reorganization of feature maps across different scales to better facilitate object detection. By leveraging the feature pyramid, the neck ensures effective utilization of both high-level semantic information and low-level details. Ultimately, the object detection task is accomplished via the head.

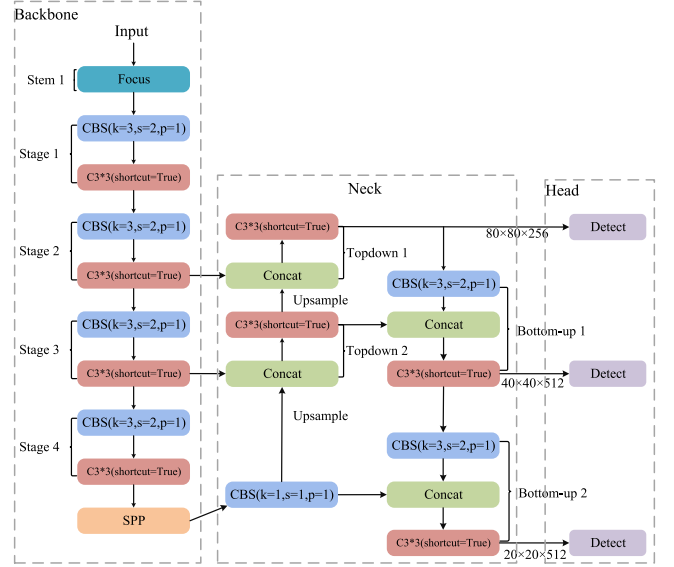


Fig. 2. Architecture of the baseline. Contains three parts: backbone, neck, and head.

TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT DETECTION ALGORITHMS ON
VISDRONE2019-TEST

Inference stage	Recall	mAP50	FPS
Without SR	43.29	38.78	72
SR	44.61	39.49	7

B. Our Method Architecture.

As shown in Fig. 3, we have two main innovations. The first major innovation involves the integration of a SR framework with our detection system. This framework is employed during the training phase to upscale low-resolution, pixelated captures into high-resolution images. By doing so, it mimics the eagle's exceptional ability to discern fine details from great distances. The SR-assisted process is meticulously designed to enhance the training data qualitatively, which in turn trains the detector to recognize finer details that are not discernible in lower resolution images. It is crucial to emphasize that the decision to apply the SR network exclusively during the training phase and omit it during inference is a strategic design choice aimed at balancing detection performance and computational efficiency. During training, the SR network enhances the quality of the input data, enabling the detection network to learn more robust and discriminative features. The network is aware of the differences between SR-assisted and unassisted images. However, incorporating SR during inference introduces significant computational overhead, which drastically reduces the frames per second (FPS), as demonstrated in Table I. This reduction in FPS would significantly impact the feasibility of real-time applications, such as drone-based object detection. Moreover, our experimental results show that the detection accuracy remains robust without SR during inference, confirming that the model generalizes well without the need for SR at this stage.

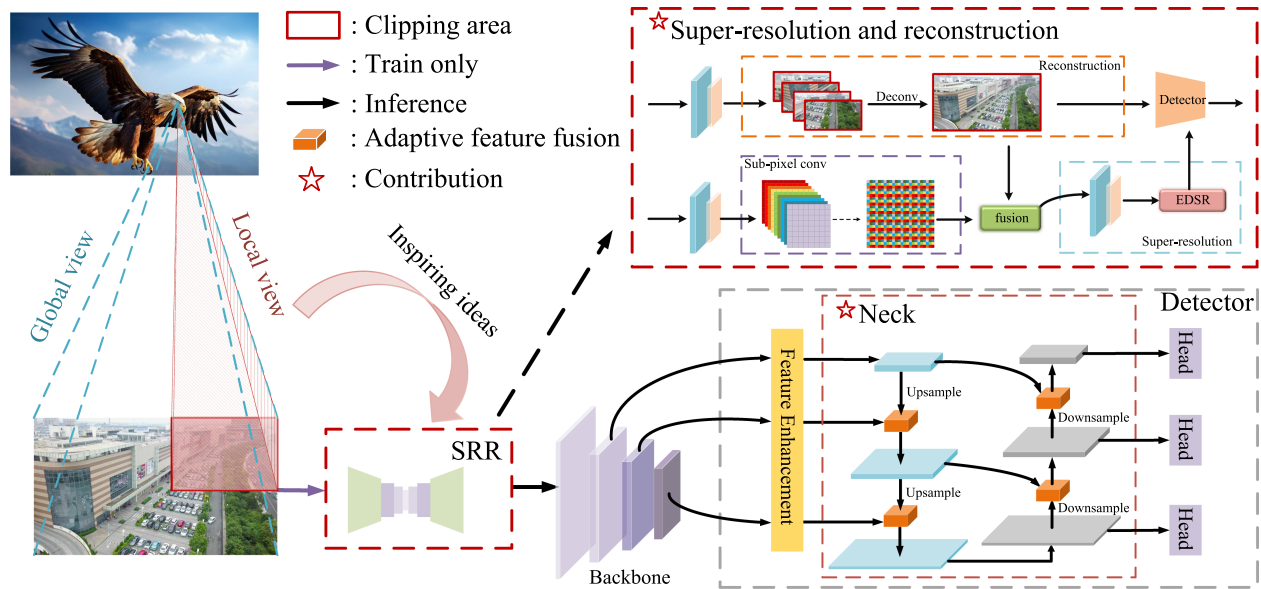


Fig. 3. Architecture of our method. Inspired by the dual field of view of an eagle's eye, the original image is cropped (with red boxes indicating the cropped areas). The cropped, localized image undergoes SR and is then reconstructed with the noncropped areas, while AFF and feature enhancement further refine the subsequent processing. Marked with a red star in the figure are the main contributions of this article. (1) The SRR model performs local SRR operations on the input image. (2) AFF evaluates the relevance and importance of features from different deep networks.

Therefore, omitting the SR network in inference optimizes the tradeoff between detection accuracy and real-time performance.

Our second innovation, the AFF mechanism, builds upon the enhanced feature set provided by the SR process. AFF method further refines the SR follow-up process by dynamically adjusting the integration of features from different layers of the network, depending on the contextual needs of the image scene. Unlike traditional feature fusion approaches that statically combine features at predetermined layers, our adaptive mechanism evaluates the relevance and importance of features from various depths of the network in real-time. The combination of the two approaches not only improves the granularity and precision of object detection but also enhances the overall robustness and speed of the detection process.

In the context of real-time UAV deployment, computational constraints are a critical consideration. Our proposed approach takes into account the limited processing power typically available on UAV systems. By omitting the SR module during the inference stage, the computational complexity is significantly reduced, which directly addresses the resource limitations of UAV platforms. This reduction in complexity results in a lower processing burden on the onboard system, thereby enhancing the real-time responsiveness of the UAV. To further quantify this computational efficiency, our experimental evaluation shows that by not utilizing the SR module during inference, the computational load—including memory and processing time—decreases substantially, making the system suitable for resource-constrained UAVs that typically have limited battery life and processing capabilities. Thus, the proposed approach ensures an optimal balance between computational efficiency and detection performance, rendering it feasible for practical real-time applications where hardware resources are restricted.

C. Superresolution and Reconstruction (SRR) Model

The SRR module operates in two primary stages during the training process, mimicking the way eagles focus on distant objects while maintaining a broad field of vision. Eagles possess a double fovea, enabling them to switch between high-resolution central vision and wider peripheral vision for detecting prey. Similarly, our SRR module enhances specific image regions (similar to an eagle's central focus) by applying SR techniques to improve detail, while ensuring the integrity of the surrounding areas (comparable to peripheral vision) through reconstruction. These stages involve the SR processing of selected image patches and their subsequent reconstruction into a full image format, ready for comprehensive analysis.

The region selection process in the SRR module prioritizes areas based on the spatial distribution of targets. Specifically, regions with a higher density of objects are given priority, as these areas often contain critical information for detection tasks. In addition, smaller or more distant objects, which are more susceptible to resolution degradation, are identified and emphasized for SR processing. Contextual information, such as the presence of structured environments, further guides the dynamic selection of regions to ensure computational resources are allocated effectively. Similar techniques exist in [31].

1) *Image Patch SR*: The SR process involves upscaling the resolution of each patch while enhancing important features and textures. As depicted in Fig. 3, the process begins with a subset of the image undergoing a subpixel convolution technique. This technique involves applying convolutional layers to the input image to extract feature maps, which are then upsampled to a higher resolution using pixel-shuffling methods. Pixel shuffling rearranges the elements of a tensor from a low-resolution grid

to a high-resolution grid, effectively increasing the spatial resolution of the image while retaining the integrity of its features. This step is crucial as it enhances the detail and sharpness of the image, preparing it for more detailed analysis.

Following the subpixel convolution process, the high-resolution images obtained are then fused with high-resolution reconstructed images. This fusion process involves integrating the upscaled images with separately reconstructed high-resolution images, which may have been processed using different techniques. The fusion is typically conducted using blending techniques that prioritize the retention of high-frequency details from both sets of images, thereby creating a composite image that features enhanced clarity and detail. This step is equally vital as it consolidates the strengths of both imaging techniques to produce superior-quality images that are more informative and accurate for further analysis.

Finally, the fused images are fed into the enhanced deep superresolution (EDSR) network for further SR processing. The EDSR network, an advanced deep learning model specifically designed for SR tasks, refines the image details further by applying a series of deep convolutional neural networks. This network adjusts and optimizes the pixel values to produce sharper, more precise images. The EDSR network is particularly effective in restoring high-resolution details from low-resolution inputs, making it ideal for final stage processing in our setup. This step is crucial as it ensures that the final output maximizes the perceptual quality of the images.

2) *Patch Reconstruction*: As shown in Fig. 3, the yellow dashed box indicates the reconstruction process. Initially, the input image is divided into smaller patches using a grid-based segmentation approach. This step mimics the focal adjustment seen in predatory birds, allowing the system to concentrate computational resources on image areas that are most likely to contain relevant information. Each patch is then independently processed through the Deconv method. This method enhances the resolution of the patches, improving the clarity and detail of features within each segment.

Following the enhancement of individual patches, they are reconstructed into their original configuration to form a complete high-resolution image. This step is crucial as it combines the detailed local insights gained from the SR process with the global context of the original image. The reconstruction algorithm ensures that the transition between enhanced patches and the surrounding image areas is seamless, maintaining the integrity of spatial relationships and minimizing any artifacts introduced during the patchwise enhancement process. The reconstructed high-resolution image is then fed into the object detection network. This part analyzes the enhanced image to identify and classify objects with improved accuracy, leveraging both the local details and overall image context. The reconstructed image will also be fused with the high-resolution image.

This dual-input strategy adopted by SRR leverages both the global contextual information from the full scene and the localized high-definition insights, markedly improving detection accuracy and recall. This method not only significantly surpasses traditional detection techniques in terms of detail and precision, but also optimally utilizes the SR enhancements to effectively

simulate an “eagle-eye” view. The focus on high-priority areas is reflected in the patch-based SRR process, ensuring that critical regions are given heightened attention. This makes it especially advantageous for UAV applications, where visual acuity from a distance is critical.

To illustrate the effectiveness of SRR more simply, we visualize the features in layers 5, 7, 9, 12, 15 of the detector, and Fig. 4 shows the effect of SRR adoption or not on the quality of the feature map. Fig. 4(a) displays partial feature heatmaps from YOLOv5s, illustrating that due to the resizing of the input image and subsequent downsampling, the extracted feature maps lack clarity, with low-resolution representations of object positions and structures. Such features are likely to have lost detailed information and characteristics of distant targets. In contrast, Fig. 4(b), which represents features processed through an SR operation, exhibits much clearer details of the targets, particularly for small and distant objects. This indicates that the high-resolution features obtained after processing through the SRR module possess clearer and more detailed information about the targets. Feature visualization reveals that AFF effectively leverages these enhanced high-resolution features to capture scale-specific details by utilizing a multiscale attention mechanism. For instance, finer details such as edges and textures of small objects, which are amplified by the SRR module, are further emphasized in the lower layers of the network, while broader semantic information for larger objects is preserved and refined in the deeper layers. Through dynamic weighting, AFF adaptively combines these features to ensure a balanced representation across scales. Visualized further demonstrate that the AFF module intensifies focus on relevant regions for small objects, such as vehicles in dense urban areas, while also maintaining the contextual integrity needed to accurately detect larger objects in the same scene.

D. Adaptive Feature Fusion

The AFF method is designed to dynamically integrate and enhance feature representations from varying dimensions and scales, ensuring optimal information synthesis for object detection in UAV-captured images. This module plays a crucial role in processing and combining the multiscale features generated in Section III-A. The fusion method is shown in Algorithm 1.

The AFF mimics the eagle’s ability to dynamically process and integrate visual information from different focal distances. Just as an eagle adjusts focus to prioritize distant prey or nearby obstacles, our AFF technique selectively enhances and fuses features from different scales and layers of the network, ensuring that both small and large objects are detected with optimal accuracy. Specifically, the feature selection criteria in AFF are guided by context-aware weighting mechanisms, which dynamically assign importance to features based on their relevance to the detection task. This mechanism evaluates spatial and semantic consistency across layers, allowing the network to prioritize features that contribute most to accurate object detection. Moreover, AFF effectively handles varying object scales by integrating high-resolution spatial details for small objects and low-resolution contextual information for larger objects. This

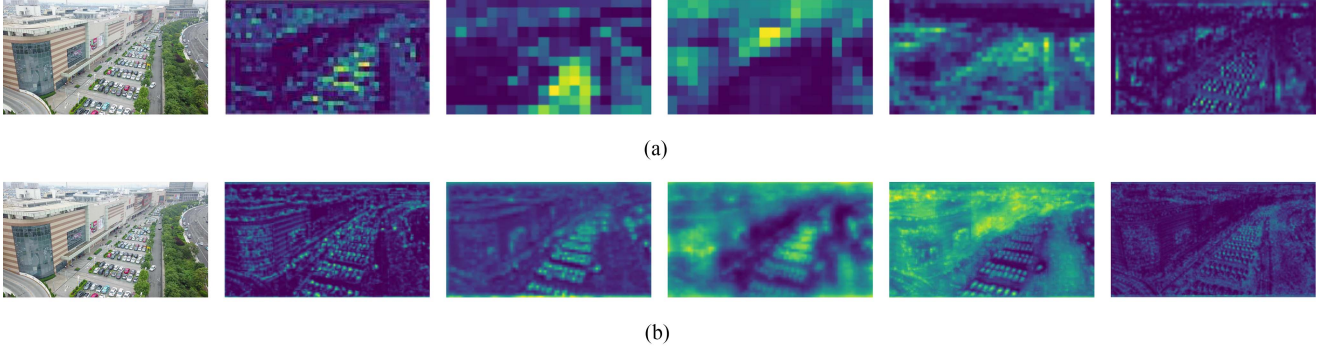


Fig. 4. Visualization of features from YOLOv5 and our method. Features were extracted from layers 5, 7, 9, 12, and 15, highlighting the impact of SRR on feature map quality. (a) Feature extraction visualization for YOLOv5. (b) Feature extraction visualization for our method.

Algorithm 1: Adaptive-Fusion Method.

Input: Multi-channel feature ($\mathcal{F}_i$)
Output: Fusion feature (F)
Fusion_layers = []
for inc in inc_list **do**
 Fusion_layers $\leftarrow$ Conv(F_i) ;
end
for $i \leftarrow 1$ **to** N **do**
 $G_i = Adaptive_Conv_layers(Concat(\mathcal{F}_i))$;
end
 $S_i = Single_graph_i = Split(SoftMax((G_i)))$
 $F = \sum_{i=1}^N Stack(S_i) * S_i$
return F

ensures robust performance across diverse scenes, particularly in UAV applications where object sizes can vary significantly within the same frame.

Initially, each input feature map, which represents different scales and dimensions as derived from the preliminary feature extraction phase (see Section III-A), is individually processed by a convolutional layer. These layers are designed to standardize the channel dimensions across different feature maps, making them amenable to uniform processing and integration. Each layer is parameterized by convolution operation fusion layers, where inc represents the number of input channels of the respective feature map, ensuring that the output maintains the same dimensionality. Following the initial convolutional processing, the AFF module employs an adaptive fusion convolution layer, *Adaptive_Conv_layers*. This layer takes the concatenated outputs of the featurewise convolutional layers and produces a set of adaptive weights. These weights are computed for each feature map to capture the relative importance of features from different levels of abstraction and dimensionality.

The output from the adaptive fusion layer undergoes a softmax operation, ensuring that the resulting weights are normalized and sum to one. This normalization is crucial for maintaining the balance in contribution across different feature maps and avoiding dominance by any single feature scale. Each normalized weight is then multiplied with its corresponding feature

map. This product operation is individually performed for all feature maps, and the results are aggregated by summation. This weighted summation effectively combines features across different scales, emphasizing those features that are deemed more relevant by the adaptive fusion process.

The AFF module leverages efficient GPU-accelerated operations for both convolution and softmax computations, ensuring that the fusion process is computationally feasible even with high-dimensional data and real-time processing requirements typical in UAV applications. The output of the AFF provides a refined feature representation that is fed into the subsequent detection head. This enhanced feature set enables the detection model to perform with increased accuracy, especially in scenarios where the differentiation of objects requires high fidelity in feature representation.

IV. EXPERIMENTS

A. Description of the Experiment

Datasets, experiments setting, and evaluation metrics are as follows.

1) *Datasets:* Two public datasets were employed for this experiment: the VisDrone2019 [32] and DOTA v1.0 datasets.

VisDrone: The dataset contains 10 209 UAV images from different cities (including Tianjin, Guangzhou, Dalian, etc.), different scenarios (e.g., city streets, rural areas, markets, etc.), and different times of the day (daytime and nighttime), with a training:validation:testing of 6471:548:1610, and an image resolution ranging from 480×360 to 2000×1500 , and a total of ten categories. In addition, these images were captured at different altitudes, ranging from low-altitude flights ideal for capturing ground-level details to higher altitude flights for broader surveillance views.

DOTA v1.0: The dataset contains 2806 aerial images from different sensors and platforms, which are widely used in remote sensing target detection. The image resolution varies from 800×800 to 2000×2000 with 15 categories. In order to train in YOLO detector, we crop the original image to 1024×1024 , and the training:validation: testing is 7:1:2 in the experiment. Similarly, the DOTA dataset includes aerial images obtained at different heights, contributing to the diversity in object sizes and scales.

2) *Experiments Setting*: All experiments were conducted on a computer running Ubuntu 16.04 as the operating system. The hardware configuration included an Intel Core i7-8700 K CPU and four NVIDIA TITAN Xp GPUs, which were utilized to accelerate the training process. Adopt PyTorch 1.12.0 framework along with torchvision 0.13.0, and torchaudio 0.12.0 to perform data training and object detection. The entire setup was supported by CUDA 10.2. For the training of our models, we employed the stochastic gradient descent optimizer with an initial learning rate of 0.01. This configuration was chosen to balance the convergence speed and training stability, enabling robust learning detection tasks.

3) *Evaluation Metrics*: We adopted recall, mAP50/50-95 as the evaluation metrics for the experiments. Recall quantifies the proportion of actual positives (true objects or conditions of interest) that are correctly identified by the model. It is calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (1)$$

precision measures the accuracy of the positive predictions made by the model. It is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

where TP denotes the true positive rate, while FP and FN denote the false positive and false negative rates, respectively.

Mean average precision (mAP) is a comprehensive metric that combines recall and precision, to evaluate the model's overall performance across multiple classes or thresholds. mAP50 indicates detection performance at a threshold of 0.5. mAP50-95 represents the average mAP calculated at multiple IoU thresholds from 0.50 to 0.95, typically at intervals of 0.05. The mAP generally expressed as follows:

$$\text{mAP} = \frac{\text{AP}_1 + \text{AP}_2 + \dots + \text{AP}_{n-1} + \text{AP}_n}{n}. \quad (3)$$

4) *Contrasting Approaches*: To demonstrate the validity of our method, the following numerous detection methods have been selected for comparison, including Cascade-Rcnn [33], FCOS [34], CenterNet [35], TridentNet [36], FSAF [37], VFNet [38], DDOD [39], TOOD [40], ATSS [41], Deformable DETR [42], YOLOv3 [43], YOLOv4 [44], YOLOX [45], YOLOv6 [46], YOLOv7 [47],¹ YOLOv8, TPH-YOLOv5 [48], CZ Det [28], RT-DETR [49], Drone-YOLO [13], HIC-YOLOv5 [50], GBS-YOLOv5 [51], MFFSODNet [17], DTSS-Net [14], Li et al. [52], Yang et al. [53], Wu et al. [15], Fovea box [54], O2D-Net [55], ARSD [56], YOLOv9 [57], YOLOv10 [58].

B. Experiment Results

Table II summarizes the comparative results of several advanced object detectors along with our proposed method, arranged in chronological order of their introduction. Compared to a subset of existing high-performing detectors, our method achieved more competitive detection performance. In the test

dataset, our approach exhibited improvements of 8.89% in recall and 7.10 in mAP50-95, showing superior performance. It performed slightly lower on the mAP50 metric compared to the method proposed by Wu et al. However, it is important to note that their model requires approximately 3.5 times more parameters and 2.5 times the computational complexity than our method. In the validation dataset, our recall was marginally lower than that of MFFSODNet, which is optimized specifically for detecting small objects and often omits detection for larger targets, contributing to its lower recall rates. Our method, on the other hand, strives for balanced detection across various object scales. Although our approach entails higher complexity and a larger number of parameters compared to the baseline model, the significant improvements in precision and recall justify these tradeoffs. It should be noted that due to limitations imposed by the output of the SR network, our experiments were conducted exclusively on the YOLOv5small (s) model size.

To further validate the effectiveness of our model, we conducted identical experiments on the DOTA v1.0 dataset. As shown in Table III, our method achieved superior detection results, particularly excelling in the recall and mAP50-95 metrics (66.38 and 40.47, respectively). In addition to general-purpose object detectors, we also considered specialized detectors designed for specific visual tasks: the fovea box detector, which is based on human visual perception, the O2D-Net tailored for RSI, and the lightweight remote sensing detector ARSD. In terms of mAP50, our detector slightly underperformed compared to O2D-Net, but the latter has a larger parameter count and increased complexity might entail higher computational costs. ARSD offers a better balance between precision and parameter efficiency, whereas our method boasts lower algorithmic complexity. Thus, our approach also presents an excellent balance between performance and cost.

The detection results at different heights of our method compared to other detectors are illustrated in Fig. 5, where we have standardized the color palettes of different models to optimize the visual clarity of the visualization. In the figure, different colored circles represent the effectiveness of the detection areas: red circles indicate missed or false detections, while green circles signify correct detections and detections beyond the ground truth (GT) annotations that were not marked but were detected. Fig. 5(a) and (c) depicts scenarios from elevated viewpoints featuring numerous small, and even minuscule, targets situated at considerable distances. In Fig. 5(a), regardless of the detection method employed—be it the baseline, YOLOv8, or RT-DETR—the performance in detecting distant small vehicles is suboptimal. Similarly, pedestrians at the entrance of a shopping center and vehicles parked nearby are not detected, with a van on the rooftop also missed. In contrast, our method not only detects distant cars and pedestrians but also identifies the easily overlooked van on the rooftop. This enhancement is attributed to the assistance of an SR network, which by elevating image quality, augments the visibility and recognizability of features associated with small targets, thereby improving the detection rate of such objects. Similarly, in Fig. 5(c), vehicles and pedestrians located in the woods and at the bottom right corner are rather inconspicuous. While RT-DETR manages to detect one vehicle, other conventional detectors largely fail.

¹[Online]. Available: <https://github.com/ultralytics/ultralytics>

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT DETECTION ALGORITHMS ON VISDRONE2019-TEST

Model	Test dataset			Val dataset			Parameters	GFLOPS
	Recall	mAP50	mAP50-95	Recall	mAP50	mAP50-95		
Cascade-Rcnn (2018)	27.51	30.54	17.87	33.50	36.07	22.53	69.39 M	355.27
FCOS (2019)	26.43	29.49	16.35	27.3	31.9	19.0	32.02 M	200.55
CenterNet (2019)	27.03	29.60	18.71	27.2	33.6	18.7	14.43 M	51.05
TridentNet (2019)	26.79	29.41	16.57	29.5	35.3	20.7	33.85 M	759.19
ATSS (2020)	30.6	32.0	18.9	32.4	36.2	22.1	32.29 M	49.22
FSAF (2019)	28.20	32.37	17.92	29.6	36.4	20.8	36.19 M	206.28
VFNet (2019)	29.94	29.55	19.86	32.2	37.3	23.1	32.50 M	189.42
DDOD (2021)	30.92	32.62	19.71	32.6	38.2	23.3	32.15 M	181.90
TOOD (2021)	31.30	33.01	20.23	33.9	39.8	24.4	31.81 M	180.95
Deformable DETR (2020)	27.53	32.13	18.75	—	—	—	39.83 M	143.69
YOLOv3 (2018)	31.91	27.11	14.75	35.67	31.9	17.69	60.62 M	99.11
YOLOv4 (2020)	32.03	26.05	14.12	37.39	32.96	17.95	52.51 M	76.34
YOLOv5s (2020)	34.40	27.67	14.69	37.56	29.73	17.09	7.08 M	10.50
YOLOv5m	34.55	28.90	15.81	41.94	37.83	21.10	21.09 M	50.42
YOLOXs (2021)	26.40	27.72	14.73	29.04	32.65	17.96	8.97 M	33.51
YOLOXm	29.8	31.5	17.1	32.4	36.6	20.7	25.33 M	92.2
YOLOv6s (2022)	31.97	31.16	18.12	35.46	36.67	22.55	16.29 M	44.1
YOLOv7s (2022)	36.42	32.88	18.08	42.68	41.69	22.67	17.76 M	48.2
TPH-YOLOv5 (2021)	36.7	—	16.6	—	—	—	9.21 M	—
RT-DETR-r18 (2023)	35.97	34.61	19.96	40.51	42.16	25.89	20.09 M	58.4
YOLOv8s (2023)	35.04	33.44	19.33	39.6	37.92	23.67	11.13 M	28.6
Li et al. (2023)	35.1	33.7	—	41.0	42.2	—	9.66 M	—
GBS-YOLOv5s (2023)	—	35.3	20.0	—	—	—	—	—
HIC-YOLOv5s (2023)	—	36.95	20.85	—	44.31	25.95	—	—
Drone-YOLOv5s (2023)	—	35.6	—	—	44.3	—	10.9 M	—
DTSSNet (2024)	—	—	—	—	41.1	—	10.1 M	—
YOLOV9s (2024)	37.83	37.17	21.62	45.39	45.17	25.91	15.59 M	67.7
YOLOV10l (2024)	37.5	37.2	20.9	46.2	45.6	26.1	25.73 M	126.4
Yang et al. (2024)	—	30.6	—	—	—	—	—	—
MFFSODNet (2024)	—	—	—	24.1	45.5	—	4.54 M	—
Wu et al. (2024)	—	42.9	—	—	—	—	62.5 M	158.8
Our	43.29	38.78	21.79	46.87	45.42	26.11	14.01 M	64.25

¹ The bold values are the best in the current column.

² The algorithms in the table are ordered according to their publication time.

However, our approach successfully identifies both vehicles, a van, and a pedestrian in these scenes. In addition, while other detection methods frequently miss most of the bicyclists at the intersection, our method is capable of recognizing motor and the cyclists riding them. This capability stems from the fact that in original low-resolution images, the absence of detail can lead detectors to either fail to distinguish between the background and small targets or erroneously recognize multiple targets as one. SR processing enhances the image's detail, thereby reducing such cases of misidentification and missed detections.

Fig. 5(b) and (d) features images captured from a lower altitude with a more horizontal perspective, thus emphasizing the challenge of multiscale aerial target detection. As demonstrated in Fig. 5(b), detectors generally perform well in identifying densely clustered cars in the upper right corner of the image. However, as the targets increase in size toward the left, the frequency of false detections escalates, with instances, such as glass reflections being misidentified as targets and vans mistakenly classified as trucks. There are also occurrences of missed

detections at the bottom of the image. Our method, in contrast, exhibits minimal issues with false and missed detections for such targets. This efficacy largely results from the capabilities of the AFF module in processing multiscale information, enabling the network to more effectively comprehend and integrate features across various layers and sizes. The AFF module dynamically adjusts the importance of each feature, ensuring effective detection regardless of the target's size. This dynamic adjustment facilitates a robust detection performance, effectively handling the diverse scales and complexities inherent in such images.

As illustrated in Fig. 5(e), the proposed method maintains robust target detection performance even in low-light conditions. Similar to detections in normal lighting, the other method does exhibit missed detections of distant small objects and more concealed objects. While other detectors are capable of identifying some motorcycles, their inherent low sensitivity to aerial targets limits their ability to effectively distinguish between motors and cyclists. In contrast, our approach, by integrating high-quality features appropriately, significantly enhances the

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT DETECTION ALGORITHMS ON DOTAv1.0

Model	Recall	mAP50	mAP50-95	parameters	GFLOPs
YOLOv3 (2018)	57.76	57.29	33.02	60.62M	196.2
YOLOv4 (2020)	57.14	59.03	33.96	52.51M	147.6
YOLOv5s (2020)	56.39	57.74	32.67	7.08M	10.53
YOLOv6s (2022)	59.42	61.17	34.26	16.29M	44.1
YOLOxs (2021)	57.62	59.32	31.16	8.94M	33.34
VFNet (2019)	53.07	59.29	31.28	32.52M	189.66
Fovea box (2020)	49.74	53.31	25.64	36.04M	203.08
TOOD (2021)	57.53	60.28	29.06	31.81M	180.95
O2D-Net (2020)	-	71.10	-	209.0M	136.04
YOLOv7s (2022)	60.36	61.23	34.32	17.76M	47.8
ARSD (2022)	-	68.28	-	13.08M	68.03
YOLOv8s (2023)	63.72	64.11	38.93	11.13M	28.5
RT-DETR-r18 (2023)	58.31	59.13	31.77	19.89M	222.9
YOLOv9s (2024)	64.62	65.30	38.13	15.59M	112.6
Our	66.38	68.15	40.47	14.01M	56.89

The bold values are the best in the current column.

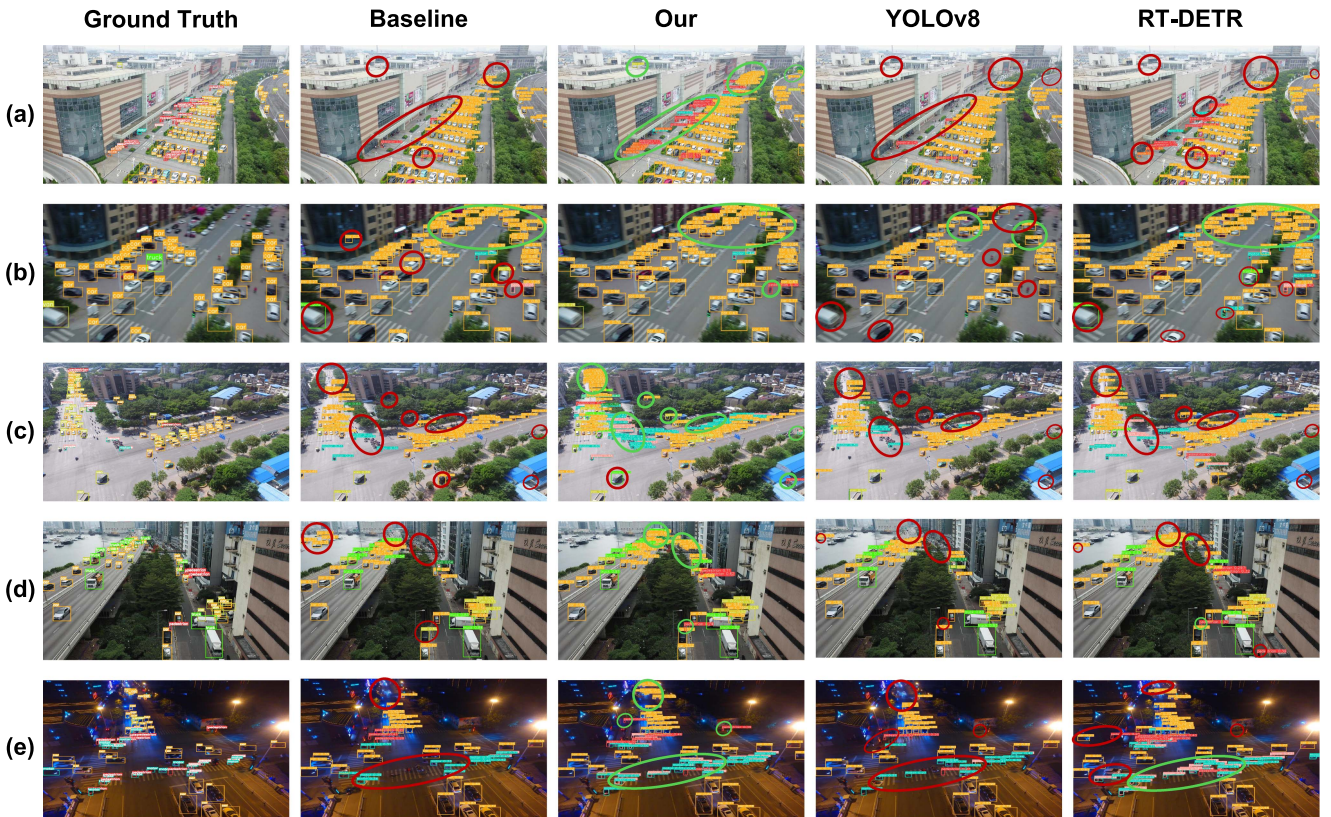


Fig. 5. Detection visualization of our method and other state-of-the-art method in VisDrone2019. Red circles indicate missed or false detections, while green circles signify correct detections and detections beyond the GT annotations that were not marked but were detected. (a) and (c) mainly contain the problem of more small objects, (b) and (d) have uneven object distribution and large scale variations, and (e) represents UAV objects at night in low light.

TABLE IV
ABLATION EXPERIMENT RESULTS ON THE VisDrone2019-TEST

SR	Reconstruction	AFF	Concat	Recall $\uparrow$	mAP50 $\uparrow$	Params $\uparrow$	GFLOPs $\uparrow$
–	–	–	✓	34.40	27.67	7.08 M	10.50
✓	–	–	✓	42.39($\uparrow$ 7.99)	38.12($\uparrow$ 10.45)	13.60 M	52.78
–	✓	–	✓	40.17($\uparrow$ 5.77)	36.50($\uparrow$ 8.83)	13.60 M	52.78
✓	✓	–	✓	42.66($\uparrow$ 8.26)	38.15($\uparrow$ 10.48)	13.60 M	52.78
✓	✓	✓	–	43.29($\uparrow$ 8.89)	38.78($\uparrow$ 11.11)	14.01 M	62.25

detection of most small and concealed targets and improves the differentiation between people and motorcycles.

The visualization of detections demonstrates that combining SR network assistance with AFF not only enhances target detection from both an image quality and feature perspective but also boosts the adaptability and efficiency of the detection system through intelligent feature processing strategies. It is noteworthy that some distant targets, which are unannotated in the GT, are typically overlooked by contemporary state-of-the-art detectors, which commonly fail to detect small objects. However, our method effectively identifies such targets, showcasing its superior detection capabilities in challenging scenarios. Our method consistently demonstrates effective detection performance across these varying altitudes, efficiently handling both the smaller, distant objects seen from higher altitudes and the larger, closer objects observed from lower altitudes.

C. Ablation Experiments

Table IV shows the results of ablation experiments of our method on VisDrone2019-test. Introducing an SR module significantly improves the capability of the network to reconstruct high-resolution details from pixelated captures, ideal for UAV imagery where capturing fine details from significant heights is crucial. This integration mimics an eagle's ability to perceive fine details from high altitudes. The addition of the SR module increased the algorithmic complexity and the parameter count, while the mAP50 metric improved by 10.45 points, illustrating the enhancement in detection capabilities for small-sized targets that traditional detection networks often miss. Furthermore, the results indicate that the reconstruction module, when utilized alone, provides a gain of 5.77 and 8.83 for a baseline on recall and mAP50 metrics, respectively. This demonstrates its effectiveness in improving image quality by optimizing pixel values across the image, yet the gains are not as significant as compared to the SR module.

However, the integration of both the SR and reconstruction modules, along with the AFF module, which dynamically adjusts the integration of features from different layers of the network depending on the contextual needs of the image scene, marked a substantial improvement over a single SR. This combination provides a gain of 8.89 and 11.11 for the baseline on recall and mAP50 metrics, respectively. The experiment results emphasize the importance of both high-resolution detail enhancement and intelligent feature integration in detecting small-sized targets, showing that a reasonable allocation of

TABLE V
PERFORMANCE COMPARISON OF YOLOv9s, YOLOv9s WITH SRR AND OTHER SR-BASED APPROACHES

Model	Recall	mAP50	Parameters
YOLOv5s	34.40	27.67	7.08 M
CZ Det	–	33.22	–
SuperYOLO	36.13	35.92	21.35 M
YOLOv9s	37.87	37.17	15.59 M
YOLOv9s+SRR	42.16	39.50	22.61 M
Our	43.29	38.78	14.01 M

learning weights according to feature importance can effectively improve detection performance.

Traditional detection networks often exhibit significant limitations when applied to UAV aerial imagery, particularly in scenarios involving high-altitude photography over large areas. First, these networks generally depend on the original resolution of the input images, which can be inherently low due to the distance of UAVs and the limitations of their cameras. This results in insufficient detail for accurately detecting small objects. Second, feature extraction in traditional networks is confined to the quality and resolution of available images, often overlooking crucial visual cues such as edges and textures, especially in low-quality images. In addition, despite being designed for computational efficiency, traditional networks may still face significant computational demands when processing large high-resolution images, impacting their real-time processing capabilities. In contrast, SR-assisted networks enhance the overall image quality, improving feature extraction and the robustness of object detection, especially in complex or varied-size object scenarios. This explains to some extent why SR techniques are a better aid to traditional detection networks on UAV aerial images.

In addition, we explored the performance of YOLOv9 in combination with our SRR approach. Based on the comparison in the Table V, the incorporation of SRR model into the YOLOv9 model significantly enhances both recall and mAP50. Specifically, the YOLOv9 combined with SRR model achieves a recall of 42.16%, a notable improvement over the baseline YOLOv9 model's recall of 37.87%. Similarly, the mAP50 increases from 37.17% to 39.50%, indicating better detection accuracy across various thresholds. This improvement in performance highlights the critical role of SRR in enhancing the model's ability to detect objects with greater precision, particularly in scenarios where image details may be less distinct or resolution is a limiting factor. However, it is also important to note that these gains

TABLE VI
RESULTS OF DIFFERENT COMBINATIONS OF UPSCALING TECHNIQUES USED WITHIN THE SRR MODULE

Interpolate	DeConv	Subpixel	Recall $\uparrow$	mAP50 $\uparrow$	Params $\downarrow$	GFLOPs $\downarrow$	Training time (min/epoch)
–	–	–	34.40	27.67	7.08 M	10.50	10
✓	–	–	40.55($\uparrow$ 6.15)	36.57($\uparrow$ 8.90)	9.96 M	64.25	52.78
✓	✓	–	41.26($\uparrow$ 6.86)	37.09($\uparrow$ 9.42)	12.29 M	64.25	52.78
✓	–	✓	42.97($\uparrow$ 8.57)	38.69($\uparrow$ 11.02)	13.32 M	64.25	52.78
✓	✓	✓	43.29($\uparrow$ 8.89)	38.78($\uparrow$ 11.11)	14.01 M	64.25	62.25

come with an increase in model complexity, as indicated by the rise in the number of parameters from 15.59 M to 22.61 M. While the increase in parameters reflects additional computational demands, the improvement in both recall and mAP50 underscores the functional importance of SRR in refining feature extraction and detection capabilities, making it a valuable enhancement in applications where high-resolution detection is crucial, such as in UAV-based imagery or other remote sensing tasks.

Our method, performs exceptionally well compared to other SR-based detection models, such as SuperYOLO and CZ Det, achieving a recall of 43.29 and a mAP50 of 38.78, significantly surpassing SuperYOLO (recall: 36.13, mAP50: 35.92) and establishing its robustness and accuracy in detecting objects across varying scales.

1) *Effect of Using Different Methods on Detection Effectiveness During the SRR.*: Table VI provided shows the results of different combinations of upscaling techniques used within the SRR module. The scaling factor used for the SR network is set to 4x. This choice was made based on the tradeoff between image quality enhancement and computational complexity. A scaling factor of 4x was selected to ensure that the superresolved images provided sufficient detail for accurate object detection while maintaining a manageable computational load during the training phase. Larger scaling factors, while potentially improving image quality further, were found to introduce diminishing returns in detection performance, and more importantly, significantly increase the training time and computational requirements. Therefore, the chosen scaling factor strikes a balance between enhancing the detection network's learning ability and keeping the training process efficient. First, the use of interpolation (bicubic) alone significantly improves both recall and mAP50 (recall increased by 6.15 points and mAP50 increased by 8.90 points), suggesting that even basic upscaling can enhance the ability of model to detect finer details. However, this approach increases the training time considerably and also bumps up the computational resources needed. Second, adding deconvolution to interpolation further improves the recall and mAP50 (0.71 and 0.52). This combination suggests that additional refinement of upscaled features through deconvolution helps in recovering more detailed information, contributing to better performance. In addition, interpolation adopted with subpixel achieves better performance improvements compared to when deconvolution is employed with interpolation. Subpixel convolution is particularly effective in enhancing image quality with a lesser increment in training time,

indicating its efficiency in handling upscaling tasks. Utilizing all three techniques yields the highest recall and mAP50 scores, underscoring the synergistic effect of combining multiple upscaling methods. Surprisingly, this approach also reduces the training time relative to using fewer methods, which might indicate more efficient feature learning and convergence during training.

The results demonstrate that combining different upscaling techniques within the SRR module not only improves the model's detection accuracy but also impacts the efficiency and computational demands of the training process. The optimal selection and combination of these techniques (interpolation, deconvolution, subpixel convolution) can significantly enhance the performance of UAV image detection systems, providing a balance between detection accuracy, computational resources, and training efficiency. The data clearly indicates that more complex combinations of upscaling methods can achieve better performance metrics without necessarily increasing training time linearly, suggesting that these methods are not just additive but multiplicative in their benefits.

V. LIMITATIONS AND FUTURE WORK

While the proposed integration of detector with SR networks and the introduction of AFF have marked significant strides in UAV image detection, this study is not without its limitations, which provide fertile ground for future research. First, the effectiveness of both SR and object detection components heavily depends on the quantity and quality of the training datasets. The current models may not perform optimally under conditions significantly different from those represented in the training data, such as extreme weather conditions or highly dynamic environments. Second, the computational complexity introduced by the SR process and the AFF algorithm increases the resource demands. This could limit the deployment of our proposed methods on UAVs with constrained computational capabilities, potentially affecting the system's real-time responsiveness. Finally, although the system has been validated on standard aerial datasets, its adaptability to varied and unforeseen environments remains a challenge. The performance in urban clutter or over densely forested areas may not meet the benchmarks set in more controlled environments.

To address the issue of computational demand, future research will focus on designing more efficient, lightweight models that maintain high accuracy while being deployable on less capable

hardware. This could involve the exploration of pruning techniques, quantization, and knowledge distillation to reduce model size and computational requirements. Improving the robustness of the system under varied operational conditions will require integrating other sensory data, such as infrared, alongside advanced data augmentation strategies. By incorporating infrared imaging, the system could enhance object detection under complex environmental conditions, such as low visibility or adverse weather. In addition, data augmentation should include simulations of adverse weather conditions, varying lighting scenarios, and highly dynamic backgrounds to train more adaptable models. Future studies also examine the transferability of the proposed techniques to other imaging modalities and domains, such as satellite imagery, maritime surveillance, or disaster management scenarios. This would involve adapting the current frameworks to handle the specific characteristics and challenges of these new domains.

VI. DISCUSSION SECTION

Adverse weather conditions, such as fog, rain, or haze, pose significant challenges for UAV-based object detection by introducing noise and reducing image clarity. While our current study focuses on stable environmental conditions, the integration of dehazing or denoising preprocessing techniques could potentially enhance the performance of our method in such scenarios. For instance, applying dehazing methods to mitigate the effects of atmospheric interference could improve the quality of input images processed by the SR module, allowing it to better amplify critical structural details, such as edges and contours. This enhancement would, in turn, enable the AFF module to more effectively prioritize relevant features across scales, even in low-visibility conditions. Such adaptations could expand the applicability of our method to challenging environments, such as disaster response during storms or surveillance in foggy terrains. Future work will explore these possibilities to systematically evaluate the performance of our approach under adverse weather conditions.

VII. CONCLUSION

This study has successfully demonstrated the implementation of a novel integrated system combining detection algorithms with SR networks, inspired by the acute visual capabilities of eagles. This integration significantly enhances the resolution and accuracy of UAV-captured images, thereby improving the precision of object detection tasks performed at varying altitudes. Our proposed AFF technique, which replaces traditional concatenation methods, has proven to be crucial in optimizing the synergy between different network layers, enhancing both the detection efficiency and robustness. The eagle-eye inspired approach allows for SR image enhancements that mimic the eagle's ability to discern fine details from great distances, thus enabling UAVs to capture detailed images even under suboptimal conditions. The AFF method dynamically integrates features across scales with a context-aware mechanism that significantly outperforms traditional feature concatenation in terms of detection accuracy and computational efficiency.

Comprehensive evaluations on aerial image datasets have established new performance benchmarks for UAV image detection, confirming the superiority of the proposed methodologies over conventional detection systems. This advancement holds significant potential for applications, such as surveillance and environmental monitoring, where enhanced image quality improves object recognition and terrain assessment. Higher resolution and reliable detection make UAVs more effective for tasks, such as disaster response, resource management, and public safety. Future work could explore the integration of other sensory data, such as infrared imaging, to enhance object detection under complex environmental conditions, such as low visibility or adverse weather. This multimodal approach could improve detection robustness and accuracy, expanding the applicability of the proposed system. In addition, further refinement of the proposed algorithm to accommodate real-time processing requirements could also increase its practical usability. We hope that the methodologies developed here will inspire further research and development, contributing to the ongoing evolution of UAV technology.

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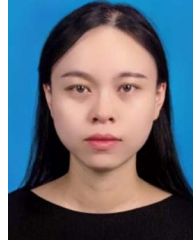
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